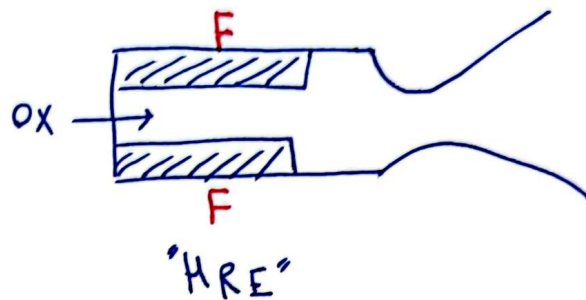


Mon-04-14-2025 "Hybrid Rocket Engines (HRE)"

(Sec. 4)

In HRE, one propellant is liquid and the other propellant is solid. Typically, the oxidizer is the liquid and the fuel is solid. HRE with liquid fuel and solid oxidizer are referred to as reverse hybrids.



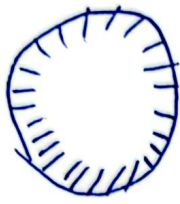
Advantages:

- ① safer handling.
- ② throttling capability (start, stop, restart)
- ③ Allows use of variety of fuels and oxidizers.
- ④ Reduced temperature sensitivity.
- ⑤ Higher I_{sp} than SRM, and higher bulk density than LPRE.
- ⑥ while cracks and voids in the solid fuel results in unpredictable operation, they are not as catastrophic as in SRM.

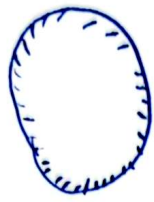


Disadvantages:

- ① oxidizer-fuel ratio changes during burn.



$t=0$



$t=t_n$

Fuel surface area (and subsequently mass) increases with time.

/// : fuel (solid)

$$r = \frac{\dot{m}_{ox}}{\dot{m}_f} \text{ changes}$$

- ② low regression rate (an order of magnitude smaller than the solid propellant). This results in complicated fuel geometry to generate significant thrust. This offsets the bulk density advantage to some extent.

- ③ lower combustion efficiency.

In the typical operating conditions of the HRE, the regression rate (r) of the fuel is empirically found to be governed by the mass flux (G), $G = \frac{\dot{m}}{A}$, i.e.,

$a', n' : \text{constants}$

$$r = a' G^{n'} \rightarrow \text{HRE}$$

$$r = a P_o^n \rightarrow \text{SRE}$$

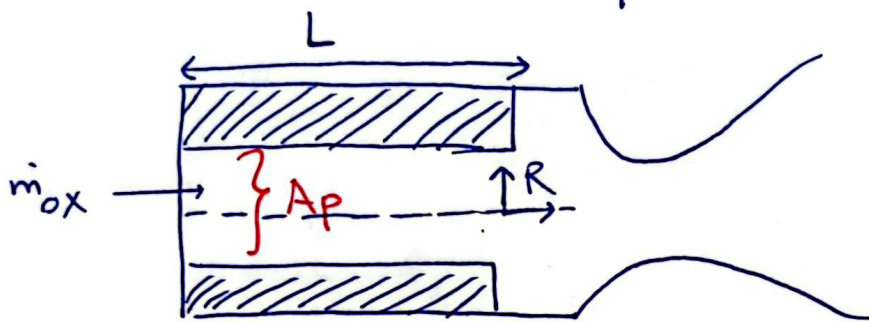
Since G includes both fuel and ox., i.e., $G = G_f + G_{ox}$, it is more convenient to use:

$$r = a G_{ox}^n \rightarrow \text{HRE}$$

$a, n \rightarrow$ empirically found.

$$\dot{m}_f = s_f r A_b = s_f a G_{ox}^n A_b \quad (A_b: \text{fuel burn area})$$

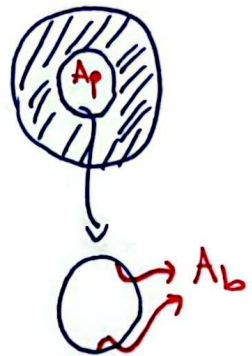
consider the simple cylindrical arrangement:



A_p : port area $A_p = \pi R^2$

$$A_b = 2\pi R L$$

$$\textcircled{1} \quad r = a G_{ox}^n = a \left(\frac{\dot{m}_{ox}}{A_p} \right)^n = a \left(\frac{\dot{m}_{ox}}{\pi R^2} \right)^n$$



$$\frac{dR}{dt} = r = a \left(\frac{\dot{m}_{ox}}{\pi R^2} \right)^n = a \frac{\dot{m}_{ox}^n}{\pi^n} R^{-2n}$$

$$\Rightarrow R^{2n} dR = a \left(\frac{\dot{m}_{ox}}{\pi} \right)^n dt$$

Integrate from $t=0 \{R=R_i\}$ to $t=t \{R=R\}$

$$\frac{R^{2n+1} - R_i^{2n+1}}{(2n+1)} = a \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t \Rightarrow$$

$$R = \left[a(2n+1) \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t + R_i^{2n+1} \right]^{\frac{1}{2n+1}} \quad (2)$$

$$\dot{m}_f = g_f A_b r = g_f (2\pi R L) a \left(\frac{\dot{m}_{ox}}{\pi R^2} \right)^n \quad (3)$$

put R from (2)

Replacing for R from (2) in (3)

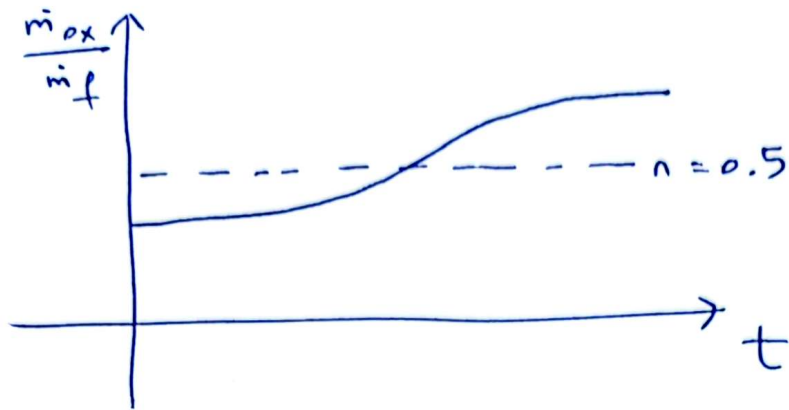
$$\dot{m}_f = 2a\pi^{1-n} g_f L \dot{m}_{ox}^n \left[a(2n+1) \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t + R_i^{2n+1} \right]^{\frac{1-2n}{1+2n}} \quad (4)$$

observe that $\dot{m}_f$ is constant if $n=0.5$

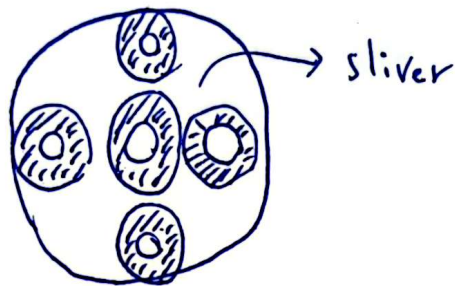
$$r = \frac{\dot{m}_{ox}}{\dot{m}_f} = \frac{1}{2g_f a L} \left(\frac{\dot{m}_{ox}}{\pi} \right)^{1-n} \left[a(2n+1) \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t + R_i^{2n+1} \right]^{\frac{2n-1}{2n+1}}$$

This is $r(O/F)$ for HRE

If $n > 0.5$, $\dot{m}_f$ decreases with time. Assuming $\dot{m}_{ox}$ is constant, $r^{O/F}$ will increase with time if $n > 0.5$



To generate large thrust, multiple ports are required, due to low regression rate of HRE. An example ($N=5$ ports) configurations is shown below.



5 ports HRE